National University of Computer and Emerging Sciences, Lahore Campus

Course:	Analog and Digital Communications	Course Code:	EE323
Program:	BS(Electrical Engineering)	Semester:	Fall 2018
Duration:	1 Hour	Total Marks:	45
Paper Date:	2-Oct-18	Weight	20%
Section:	ALL	Page(s):	3
Exam:	MID-I	Roll No:	[redacted]

Instruction/Notes: Attempt all parts of a question together.
Fourier Transforms Table is provided

Question # 1 [CLO: 1] **[15]**

A low-pass filter transfer function $H(\omega)$ is given by

$$H(\omega) = \tau\left[\frac{\sin\left(\frac{\omega\tau}{2}\right)}{\frac{\omega\tau}{2}}\right]e^{-j5\omega}$$

a. Is this a distortion-less system, if yes explain, if no then identify the type of distortion introduced by this system.

b. A signal $g(t) = \delta(t)$ is applied at the input of this filter. Find the output $y(t)$ and describe the effect of channel on input signal.

Question # 2 [CLO: 2] **[15]**

Given the signal

$$g(t) = e^{-t}u(t)$$

a) Show that $g(t)$ is an energy signal.

b) Find the autocorrelation function $\psi_g(\tau)$ of $g(t)$.

c) Show how from $\psi_g(\tau)$ you can deduce the energy spectral density $\Psi_g(\omega)$ of $g(t)$.

d) Find the ESD of output signal $\Psi_y(\omega)$ if the signal $g(t)$ is passed through a filter whose frequency response is $H(\omega) = e^{-j\omega T}$, where T is a constant.

e) Find the output signal energy E_y using output energy spectral density $\Psi_y(\omega)$ found in part d.

Question # 3 [CLO: 3] **[15]**

The spectrum of the message signal $m(t)$ is shown in Fig. 1 below.

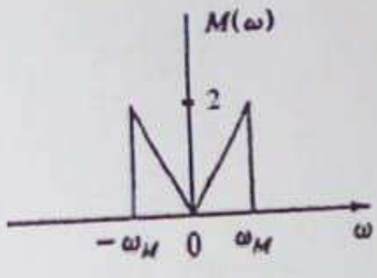

Figure 1 Spectrum of $m(t)$

a) The signal $m(t)$ is applied to a system shown in Fig 2. Analyze the system and sketch the spectrum at points A and B. Properly label the axis information. Determine the bandwidth of signals at A and B.

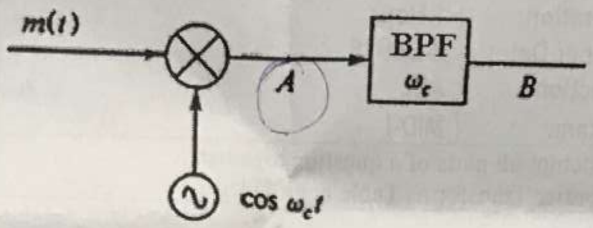

Figure 2

b) The signal $m(t)$ is applied to a system shown in Fig 3. Analyze the system and sketch the spectrum of output $x(t)$. Properly label the axis information. Determine the bandwidth of signal $x(t)$.

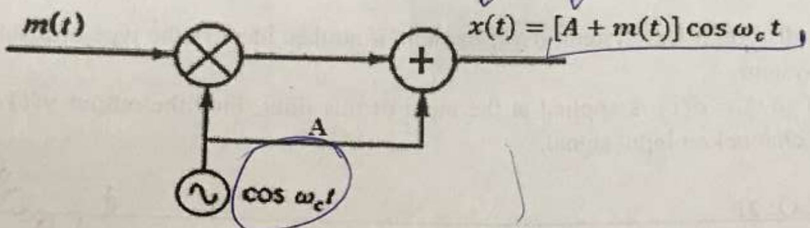

Figure 3

Note: $\omega_c \gg \omega_M$

$m(t)\cos\omega_c t$

$\frac{1}{2}\left(M((\omega+\omega_0)\right)$ -

$M(\omega-\omega_0)$